



CS2009
Spring 2024
Assignment#03

Design and Analysis of Algorithm

SUBMISSION GUIDELINES:

1. The assignment must be submitted within the due date.
2. The assignment must be on paper and in handwritten form.
3. **Plagiarism is not allowed. In the case of a pledge, you will score zero marks.**
4. Do not submit your assignment after the deadline.
5. Do not copy from any other source.
6. It is an individual submission.

Question no: 1

Scenario:

You are a financial analyst tasked with optimizing the buying and selling strategy for a particular stock. You have been provided with an array representing the daily changes in stock prices over a certain period. Your goal is to determine the contiguous subarray of days where buying low and selling high would yield the maximum profit. However, you need to choose between different techniques to solve this problem efficiently.

Given Array: [-1, 3, -2, 4, -1, 2, 1, -5, 4]

Tasks:

Marks [5]

1. Using the brute-force approach, determine the maximum profit that can be obtained by buying low and selling high. Show your step-by-step calculation and the subarray with the maximum sum.

Marks [5]

2. Implement Kadane's Algorithm to find the maximum subarray sum and determine the optimal buying and selling days for maximum profit. Show your step-by-step calculation and the subarray with the maximum sum.

Marks [5]

3. Apply the divide-and-conquer approach to solve the problem. Divide the array into subarrays, find the maximum subarray sum in each subarray, and combine the results to determine the maximum profit. Show your step-by-step calculation and the subarray with the maximum sum.

Marks [5]

4. Analyze the time complexity of each technique used in solving the problem. Compare the efficiency of the brute-force approach, Kadane's Algorithm, and the divide-and-conquer approach in terms of time complexity and practical execution time.

Marks [5]

5. Based on your analysis, discuss the advantages and disadvantages of each technique and provide recommendations for choosing the most appropriate approach for similar scenarios in the future.

Question no: 2

Marks [10]

Given the array, the objective is to find the maximum sum subarray from a given array, employing the most fitting technique. Additionally, provide reasoning for selecting this particular technique over others.

Array: [7, -2, 14, -7, 4, -3, 5, 0]

Question no: 3

Scenario:

You are a software engineer developing a text processing system for a large-scale document analysis project. Your task is to implement a string pattern matching algorithm to efficiently search for specific keywords within the text documents. However, you need to choose between different techniques, including the Naive Approach, Knuth-Morris-Pratt (KMP) Algorithm, and Rabin-Karp Algorithm, to ensure optimal performance for this task.

Given Array:

Text: "ababcababcbabcabc"

Pattern: "ababcbabc"

Tasks:

Marks [5]

1. Using the Naive Approach, perform string pattern matching to determine if the pattern exists within the given text. Show your step-by-step calculation and identify the positions where the pattern occurs.

Marks [5]

2. Implement the Knuth-Morris-Pratt (KMP) Algorithm to search for the pattern within the text. Show your step-by-step calculation and identify the positions where the pattern occurs.

Marks [5]

3. Apply the Rabin-Karp Algorithm to search for the pattern within the text. Show your step-by-step calculation and identify the positions where the pattern occurs.

Marks [5]

4. Analyze the time complexity of each technique used in solving the string pattern matching problem. Compare the efficiency of the Naive Approach, KMP Algorithm, and Rabin-Karp Algorithm in terms of time complexity and practical execution time.

Marks [5]

5. Based on your analysis, discuss the advantages and disadvantages of each technique and provide recommendations for choosing the most appropriate approach for similar scenarios in the future.

Question no: 4

Marks [10]

You are developing a plagiarism detection system for a university. The system needs to efficiently compare a large collection of student essays against a database of reference documents to identify any instances of plagiarism. Choose between different string pattern matching techniques, including the Naive Approach, Knuth-Morris-Pratt (KMP) Algorithm, and Rabin-Karp Algorithm, to ensure accurate and timely detection of plagiarized content. Explain why you choose the algorithm and why it is better than other.

Question no: 5

Marks [10]

You are tasked with implementing a spell-checking feature for a word processing application. The spell-checker needs to efficiently identify misspelled words within a document and suggest corrections. Discuss the suitability of using the Naive Approach, Knuth-Morris-Pratt (KMP) Algorithm, and Rabin-Karp Algorithm for this task. Consider factors such as time complexity, space complexity, handling of multiple patterns, and adaptability to different scenarios. Provide rationale for selecting the most appropriate algorithm for implementing the spell-checking feature and explain how it addresses the specific requirements of the task.